



Standard Form for Per-Unit-Length RLGC Matrices of Multiconductor Transmission Lines

Theory, Mathematical Foundations, Complete Stamp Catalog, and Simulation Algorithms

Abstract

This document presents the standard matrix formulation for the per-unit-length (PUL) parameters of multiconductor transmission lines (MTLs). The information presented here is based on the contents of the textbook: Clayton R. Paul's *Analysis of Multiconductor Transmission Lines* (1994) [1]. For a uniform $(n + 1)$ -conductor transmission line consisting of n active signal conductors and a common reference conductor which is the conductor numbered 0 and is also referred to as ground, the coupled Telegrapher equations describe the spatial and temporal distribution of voltages and currents along the length of the line, in the Transverse Electromagnetic Mode (TEM) or quasi-TEM regime. The transmission line dynamics are governed by four $n \times n$ parameter matrices: the inductance matrix $\mathbf{L}$, resistance matrix $\mathbf{R}$, conductance matrix $\mathbf{G}$, and capacitance matrix $\mathbf{C}$. The governing equations, physical interpretations, and standard matrix forms are presented below.

1. Multiconductor Transmission Line (MTL) System

Consider a uniform multiconductor transmission line consisting of n active signal conductors and an additional conductor, designated as conductor 0, that serves as the common reference (return) conductor [1]. The transmission line geometry and surrounding dielectric medium are assumed uniform along the longitudinal propagation axis (z).

The line voltages of the n signal conductors with respect to the reference conductor are represented by the $n \times 1$ voltage vector $\mathbf{V}(z, t)$, and the signal conductor currents are represented by the $n \times 1$ current vector $\mathbf{I}(z, t)$. These are defined in (1) as

$$\mathbf{V}(z, t) = \begin{bmatrix} v_1(z, t) \\ v_2(z, t) \\ \vdots \\ v_n(z, t) \end{bmatrix}, \quad \mathbf{I}(z, t) = \begin{bmatrix} i_1(z, t) \\ i_2(z, t) \\ \vdots \\ i_n(z, t) \end{bmatrix} \quad (1)$$

The total return current flowing through the reference conductor is given by Kirchhoff's Current Law:

$$i_0(z, t) = - \sum_{i=1}^n i_i(z, t)$$

2. Coupled Multiconductor Telegrapher Equations

The spatial distribution and temporal dynamics of the voltage and current vectors along the multiconductor line are governed by the coupled Telegrapher equations in matrix differential form [1]:

$$\frac{\partial \mathbf{V}(z, t)}{\partial z} = -\mathbf{R} \mathbf{I}(z, t) - \mathbf{L} \frac{\partial \mathbf{I}(z, t)}{\partial t} \quad (2a)$$

$$\frac{\partial \mathbf{I}(z, t)}{\partial z} = -\mathbf{G} \mathbf{V}(z, t) - \mathbf{C} \frac{\partial \mathbf{V}(z, t)}{\partial t} \quad (2b)$$

In the frequency domain (assuming sinusoidal steady-state excitation with angular frequency ω and phasor representations $\hat{\mathbf{V}}(z)$ and $\hat{\mathbf{I}}(z)$), the coupled equations become:

$$\frac{d\hat{\mathbf{V}}(z)}{dz} = -\mathbf{Z} \hat{\mathbf{I}}(z) = -(\mathbf{R} + j\omega \mathbf{L}) \hat{\mathbf{I}}(z) \quad (3a)$$

$$\frac{d\hat{\mathbf{I}}(z)}{dz} = -\mathbf{Y} \hat{\mathbf{V}}(z) = -(\mathbf{G} + j\omega \mathbf{C}) \hat{\mathbf{V}}(z) \quad (3b)$$

where $\mathbf{Z} = \mathbf{R} + j\omega \mathbf{L}$ is the $n \times n$ per-unit-length impedance matrix modeling longitudinal series loop impedance, and $\mathbf{Y} = \mathbf{G} + j\omega \mathbf{C}$ is the $n \times n$ per-unit-length admittance matrix modeling transverse shunt coupling.

3. Longitudinal Parameter Matrices ($\mathbf{L}$ and $\mathbf{R}$)

The longitudinal parameters model the magnetic flux linkages and ohmic losses associated with current loops formed by the signal conductors and the reference conductor:

- **Per-Unit-Length Inductance Matrix, $\mathbf{L}$** ($n \times n$, symmetric, $\mathbf{L} = \mathbf{L}^T > 0$):

The inductance matrix contains the individual per-unit-length self-inductances, l_{ii} , of the circuits formed between conductor i and reference conductor 0, and the per-unit-length mutual inductances, $l_{ij} = l_{ji}$, between circuits i and j , as

$$\mathbf{L} = \begin{bmatrix} l_{11} & l_{12} & \cdots & l_{1n} \\ l_{21} & l_{22} & \cdots & l_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix} \quad (4)$$

- **Per-Unit-Length Resistance Matrix, $\mathbf{R}$** ($n \times n$, symmetric, $\mathbf{R} = \mathbf{R}^T > 0$):

The resistance matrix accounts for the finite conductivity of the conductors. The diagonal entries ($r_i + r_0$) represent the total round-trip loop resistance for circuit i consisting of signal conductor resistance r_i and the common return conductor resistance r_0 . The off-diagonal entries r_0 represent the common-impedance resistive coupling due to shared return current in conductor 0:

$$\mathbf{R} = \begin{bmatrix} (r_1 + r_0) & r_0 & \cdots & r_0 \\ r_0 & (r_2 + r_0) & \cdots & r_0 \\ \vdots & \vdots & \ddots & \vdots \\ r_0 & r_0 & \cdots & (r_n + r_0) \end{bmatrix} \quad (5)$$

where $\mathbf{V}$ and $\mathbf{I}$ are defined by (1).

4. Transverse Parameter Matrices (G and C)

The transverse parameters represent the conduction and displacement currents flowing across the dielectric medium in the transverse cross-sectional plane:

- **Per-Unit-Length Conductance Matrix, $\mathbf{G}$** ($n \times n$, symmetric, hyperdominant):

Represents the transverse conduction current (dielectric leakage losses) between conductors and is defined from (2.21) as

$$\mathbf{G} = \begin{bmatrix} \sum_{k=1}^n g_{1k} & -g_{12} & \cdots & -g_{1n} \\ -g_{21} & \sum_{k=1}^n g_{2k} & \cdots & -g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -g_{n1} & -g_{n2} & \cdots & \sum_{k=1}^n g_{nk} \end{bmatrix} \quad (6)$$

- **Per-Unit-Length Capacitance Matrix, $\mathbf{C}$** ($n \times n$, symmetric, hyperdominant, $\mathbf{C} > 0$):

Represents the transverse displacement current (electric field coupling) between conductors as

$$\mathbf{C} = \begin{bmatrix} \sum_{k=1}^n c_{1k} & -c_{12} & \cdots & -c_{1n} \\ -c_{21} & \sum_{k=1}^n c_{2k} & \cdots & -c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -c_{n1} & -c_{n2} & \cdots & \sum_{k=1}^n c_{nk} \end{bmatrix} \quad (7)$$

Note on Conductor Indexing and Diagonal Summations [1]:

In Paul's formulation, conductor 0 denotes the reference (ground) conductor. On the diagonal entries of $\mathbf{G}$ and $\mathbf{C}$, the terms $g_{ii} \equiv g_{i0}$ and $c_{ii} \equiv c_{i0}$ represent parameters from conductor i directly to the reference conductor:

$$\sum_{k=1}^n g_{ik} = g_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n g_{ik}, \quad \sum_{k=1}^n c_{ik} = c_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n c_{ik}$$

while off-diagonal terms ($-g_{ij}$ and $-c_{ij}$, for $i \neq j$) model mutual transverse coupling between signal conductors i and j .

References

- [1] C. R. Paul, *Analysis of Multiconductor Transmission Lines*. New York, NY, USA: John Wiley & Sons, Inc., 1994, ch. 2, pp. 51–55.